Chapter 10

Background

The electrographic background reflects the patient's state and level of consciousness.

You can describe it based on the following six features:

1. symmetry
2. continuity
3. voltage
4. organization (anterior–posterior gradient, predominant frequencies, and posterior-dominant rhythm)
5. reactivity (or variability)
6. sleep architecture

Let us understand the usage of each term with its clinical significance.

Symmetry

To determine if the EEG background is symmetric over both hemispheres, *contrast their frequencies and amplitudes.*

First, use a longitudinal bipolar montage to compare the hemispheric frequencies and then use a referential montage to compare their amplitudes [1].

The background may be

1. symmetric
2. mildly asymmetric (if there is frequency asymmetry of less than 1 Hz or consistent amplitude asymmetry of less than 50%)
3. markedly asymmetric (if there is frequency asymmetry of more than 1 Hz or consistent amplitude asymmetry of more than 50%)

The normal background is symmetric, a mild physiological asymmetry with the dominant hemisphere having a slightly lower amplitude than the non-dominant side is within normal limits. Figure 10.1 shows a markedly asymmetric left hemispheric background.

https://doi.org/10.1017/9781108918923.012 Published online by Cambridge University Press

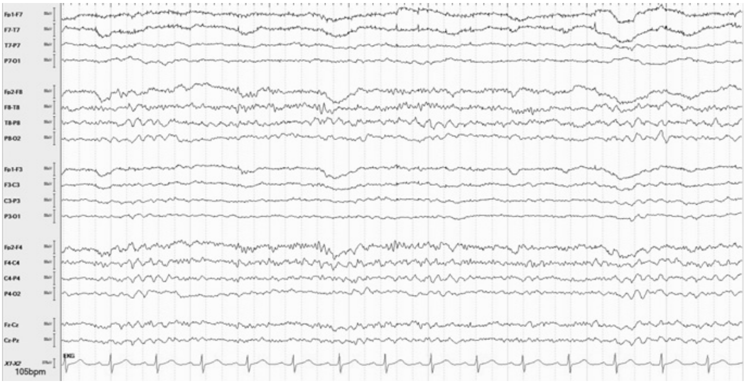

Figure 10.1 Seventy-one-year-old woman with left cerebral hemi atrophy, leptomeningeal thickening, and gyral calcifications from Sturge–Weber syndrome. There is marked asymmetry both in amplitude and frequency over the left hemisphere.

Significance

1. There is physiological asymmetry. The nondominant hemisphere (commonly right) may have a slightly higher background amplitude compared to the dominant hemisphere (commonly left). Conversely, a higher dominant background or a markedly increased nondominant background is abnormal [2].
2. Skull defects (e.g., craniotomy) leads to higher amplitudes and the presence of faster frequencies over the region, underlying activity may also appear sharper. This is called a breach effect [3]. Figure 10.2 shows a left temporal breach effect.
3. Subdural lesions (e.g., hematoma or fluid collections) predominantly dampen the amplitude of cerebral activity, they may also cause slowing.
4. Structural lesions (e.g., strokes or tumors) predominantly cause slowing of cerebral activity, they may also dampen amplitude.
5. Ictal activity may be associated with faster frequencies and higher amplitudes, while slowing and/or suppression is typical of postictal states.
6. A combination of higher amplitudes but slower frequencies may be the result of a breach effect with an underlying structural lesion (e.g., craniotomy for tumor).

If the background is markedly asymmetric, other features such as continuity, voltage, reactivity (or variability), and sleep architecture should be described separately for each hemisphere [2].

https://doi.org/10.1017/9781108918923.012 Published online by Cambridge University Press

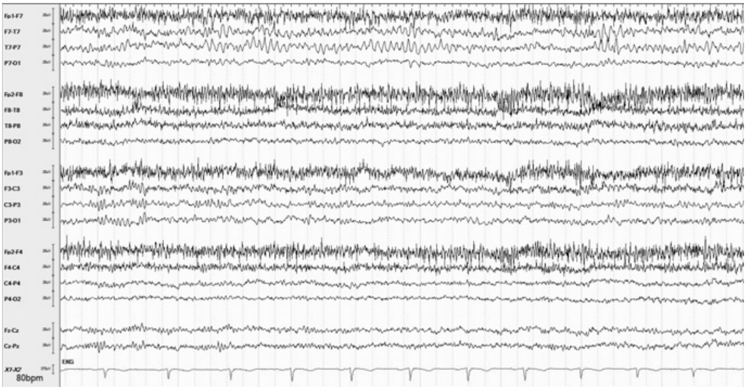

Figure 10.2 Sixty-two-year-old man with a left craniotomy for evacuation of a subdural hematoma. There is breach effect over the left temporal region (T7 channel).

Continuity

This refers to *periods of attenuation or suppression.*

Attenuation is the periodic dampening of amplitudes to less than half of the remaining background but remaining greater than 10 uV.

Suppression is severe attenuation, where amplitude of cerebral activity is less than 10 uV.

The background may be

1. continuous – without attenuation/suppression
2. nearly continuous – occasional, very brief periods of attenuation/suppression
3. discontinuous – frequent periods of attenuation/suppression that make up less than half the recording
4. burst-suppressed (or burst-attenuated), periods of attenuation/suppression for more than half of record:

   - always specify the typical duration of the bursts and the interburst intervals
   - also specify if the bursts are symmetric, synchronous and identical or nonidentical in appearance [1]

The normal background is continuous. Figures 10.3 (nearly continuous), 10.4 (discontinuous), and 10.5 (burst-suppression) show increasing levels of background discontinuity.

https://doi.org/10.1017/9781108918923.012 Published online by Cambridge University Press

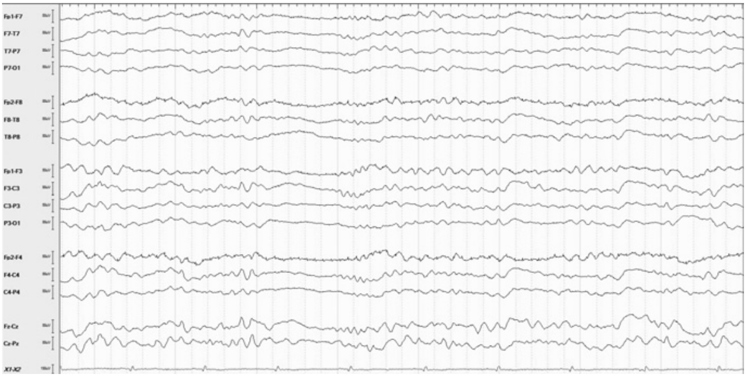

Figure 10.3 Forty-five-year-old woman sedated for episodes of agitation showing a near-continuous background. Note the brief (1 s) period of diffuse background suppression.

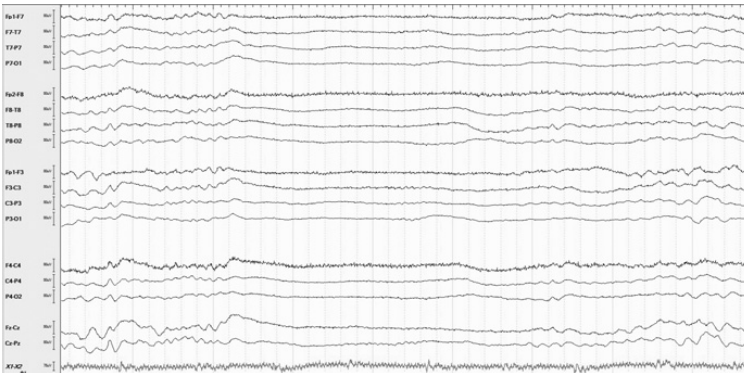

Figure 10.4 Discontinuous background with increasing sedation in the same patient as in Figure 10.3.

Significance

1. Discontinuity is an electrographic marker of the severity of encephalopathy. Longer interburst intervals (with shorter bursts) are associated with a greater depth of encephalopathy. It is nonspecific to etiology

https://doi.org/10.1017/9781108918923.012 Published online by Cambridge University Press

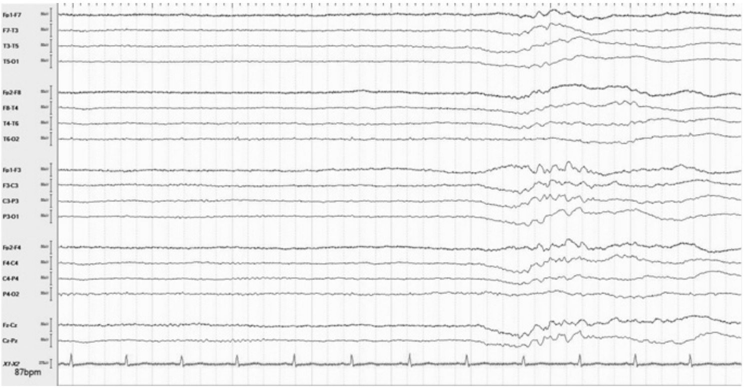

Figure 10.5 Burst-suppressed background with increased sedation in the same patient as in Figure 10.3.

but commonly associated with the use of sedative agents, metabolic disturbances (typically reversible) and in postanoxic states (typically irreversible) [4].

2. Highly epileptiform bursts (HEBs) are bursts with multiple epileptiform abnormalities (spike or sharp waves). These may be associated with ongoing underlying ictal activity such as partially suppressed or refractory status epilepticus [1,5]. Figure 10.6 shows HEBs in a patient being treated for generalized status epilepticus.
3. Burst-suppression with identical bursts is a distinct EEG pattern associated with poor outcomes in post anoxic coma. Here almost all the bursts in the record look stereotypical or identical to one another [6].

Voltage

The overall voltage or amplitude of the background cerebral activity is also called "development."

Determine the background voltage by measuring the height of the activity from peak to trough on a longitudinal bipolar montage.

Normal cerebral background voltage is variable, but greater than 20 uV (usually 30–40 uV). A low-voltage background is less than 20 uV and a suppressed background is less than 10 uV [1].

If the background is discontinuous (e.g., burst-suppressed), the background voltage refers to the amplitude of bursts [1]. Figure 10.7 shows a suppressed background.

https://doi.org/10.1017/9781108918923.012 Published online by Cambridge University Press

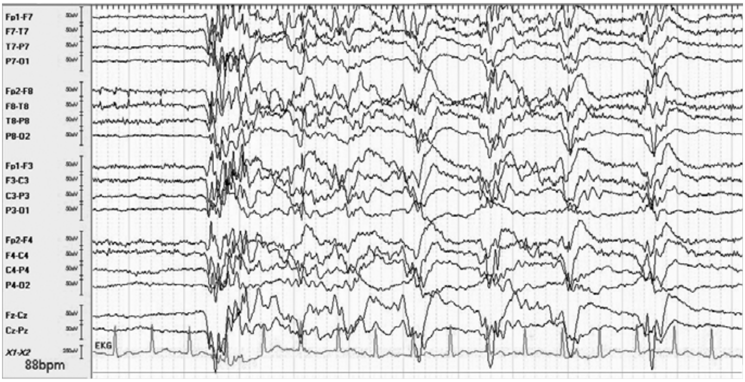

Figure 10.6 Thirty-five-year-old man being treated for generalized status epilepticus post cardiac arrest. Highly epileptiform bursts (HEBs) consisting of generalized discharges seen within bursts.

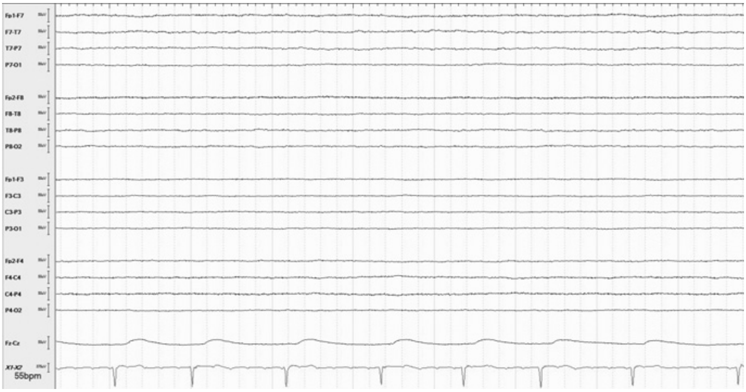

Figure 10.7 Sixty-eight-year-old woman post cardiac arrest showing a suppressed background.

Significance

1. Low background voltage is a nonspecific electrographic marker of diffuse cortical dysfunction. This may result from several different etiologies including hypoperfusion, alcoholism, medications, and metabolic

https://doi.org/10.1017/9781108918923.012 Published online by Cambridge University Press

disturbances and neurodegeneration (e.g., dementia). Occasionally, it may be entirely normal.
2. The depth of suppression correlates with the severity of the encephalopathy. Complete suppression is typical with anesthetic or post anoxic coma [7].

Organization

This is the overall layout of the background cerebral activity; it may be further described based on the following three features:

1. *Anterior–posterior gradient:* Look for an anterior to posterior gradient of voltages and frequencies such that the lower amplitudes and faster frequencies are seen in the anterior channels and the higher amplitudes and slower frequencies are seen in the posterior channels. A normal, well-organized background has an appreciable anterior–posterior gradient [1].
2. *Predominant frequencies:* Specify the typical frequency of the background cerebral activity (usually alpha, theta, or delta) during maximal wakefulness, which may be after stimulation in a poorly responsive patient. Other frequencies such as beta may also be present (e.g., moderate or excess beta is present symmetrically). The predominant background frequency during normal wakefulness is alpha.
3. *Posterior-dominant rhythm (PDR):* If present, it should be described. Normally, an alpha frequency posterior-dominant rhythm is seen in the waking state that is reactive to eye closure [1,2].

A well-organized background (normal) is shown in Figure 10.8. Note the distinct anterior–posterior gradient, predominant alpha frequency, and a reactive posterior-dominant rhythm of 10 Hz. Figure 10.9 shows a poorly organized background with loss of anterior–posterior gradient, predominantly delta and theta slowing and absence of posterior-dominant rhythm.

Significance

1. Loss of an anterior–posterior gradient, predominantly slower frequencies (theta or delta), and the absence of a posterior-dominant rhythm constitute a deterioration of background organization.
2. Poorly organized background is a nonspecific electrographic marker of the severity of cerebral dysfunction. A return of background organization correlates with the resolution of cerebral dysfunction.

Reactivity (or Variability)

Reactivity refers to changes of background cerebral activity (either amplitude or frequency) in response to external stimulation. Stimulation should be specified and advanced in a graded fashion (auditory, such as name-calling,

https://doi.org/10.1017/9781108918923.012 Published online by Cambridge University Press

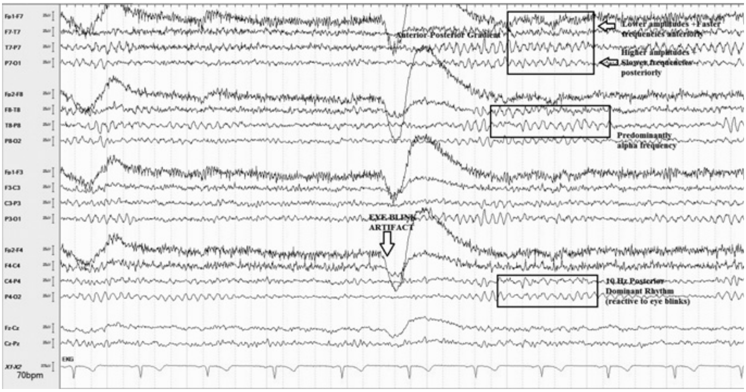

Figure 10.8 Forty-five-year-old man with episodic tinnitus showing features of a well-organized background.

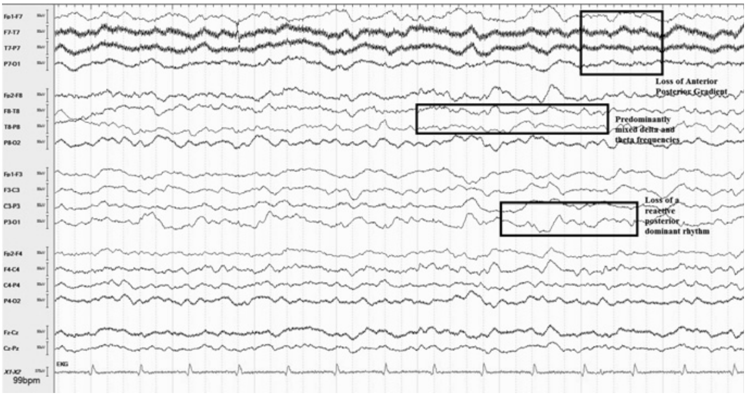

Figure 10.9 Fifty-six-year-old woman with hepatic encephalopathy showing loss of background organization.

to noxious, such as nail bed pressure). Stimulation induced foreground activities may also emerge [1]. If reactivity has not been tested, or is absent – the reader should assess the record for variability.

Variability refers to the natural transformations of background cerebral activity with fluctuations in the level of alertness or arousals. These tend to be

https://doi.org/10.1017/9781108918923.012 Published online by Cambridge University Press

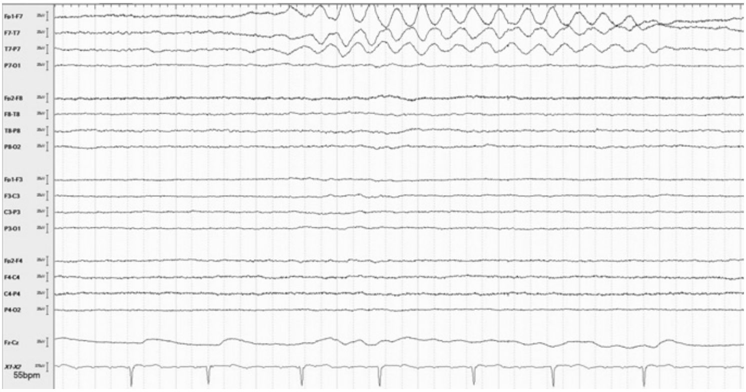

Figure 10.10 Seventy-two-year-old woman in a coma showing a rhythmic left temporal artifact upon sternal stimulation.

spontaneous (without stimulation) and sustained (at least a minute in duration). Variability is best assessed during silent periods and care should be taken to limit artifacts (e.g., respirators or IV pumps).

Normal background is reactive (e.g., eye blinks) and variable (e.g., state change). Figure 10.10 shows a sternal rub artifact over the left temporal channels, this action elicits a low-voltage burst of diffuse mixed delta and theta slowing in an otherwise suppressed background, as shown in Figure 10.11.

Significance

Reactivity (or variability) is typically associated with favorable clinical outcomes irrespective of etiology but the lack of a standardized approach to assessment and interpretation can be a limitation [7].

Sleep Architecture

The reader should assess the record for the presence of stage II sleep architecture (sleep spindles and/or K complexes). These constitute further evidence of physiological state changes. Absent or asymmetric stage II sleep architecture is usually abnormal. Figure 10.12 shows symmetric sleep spindles in a sedated patient.

Significance

1. The presence of normal sleep architecture is typically regarded as a favorable prognostic marker, irrespective of etiology [8].
2. Asymmetric K complexes or spindles may be seen with cerebral dysfunction or damage [9].

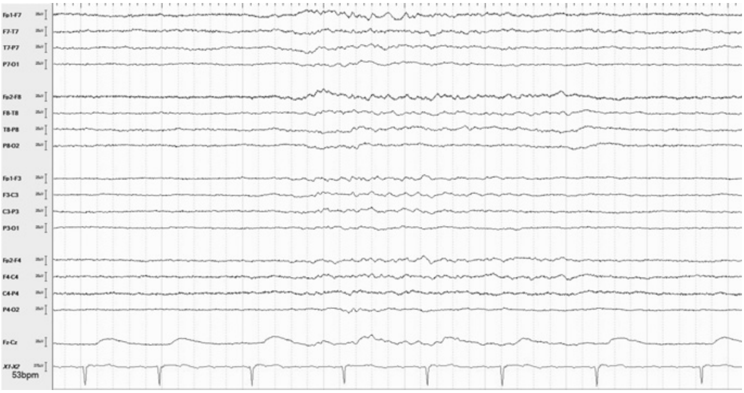

Figure 10.11 Sternal stimulation is shortly followed by a brief burst of low-amplitude activity in the same patient (Figure 10.10) indicating reactivity.

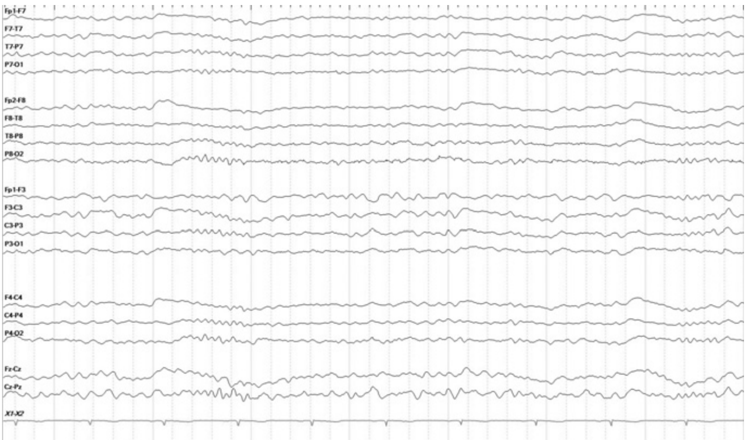

Figure 10.12 Fifty-eight-year-old woman in a propofol-induced coma showing symmetric spindles.

3. Spindle coma refers to a nonspecific electrographic pattern characterized by an abundance of spindle activity and other sleep transients in comatose individuals [8].
4. Anesthetic agents such as propofol may induce spindle activity [10].

https://doi.org/10.1017/9781108918923.012 Published online by Cambridge University Press

Chapter Summary

You can describe the electrographic background described based on the following features:

1. *Symmetry:* It may be symmetric, mildly asymmetric, or markedly asymmetric. Mild asymmetry of the dominant hemisphere may be physiologic.
2. *Continuity:* It may be continuous, nearly continuous, discontinuous, or burst-suppressed. Attenuation is the periodic dampening of amplitudes by less than half the remaining background while suppression is when the amplitude falls below 10 uV.
3. *Voltage:* It may be normal (above 20 uV), low voltage (20–10 uV), or suppressed (less than 10 uV).
4. *Organization:* A well-organized background has an anterior–posterior gradient, predominantly alpha frequencies, and a reactive posterior-dominant rhythm (PDR). The loss of either or all these features results in a deterioration in background organization.
5. *Reactivity (or variability):* Reactivity refers to changes in background amplitude or frequency in response to external stimulation, while variability refers to spontaneous fluctuations in the patient's level of alertness. Always specify the stimulation used to test reactivity and advance it in a graded manner. Reactivity (or variability) of the background are associated with favorable clinical outcomes.
6. *Presence of stage II sleep architecture such as spindles and K complexes should be noted:* They may be symmetric or asymmetric. Sleep spindles may be induced by anesthetics such as propofol. Sleep spindles and K complexes are associated with favorable clinical outcomes.

References

1. Hirsch LJ, LaRoche SM, Gaspard N, et al. American clinical neurophysiology society's standardized critical care EEG terminology: 2012 version. *Journal of Clinical Neurophysiology*. 2013 Feb 1;**30**(1):1–27.

2. Britton JW, Frey LC, Hopp JL, et al. *Electroencephalography (EEG): an introductory text and atlas of normal and abnormal findings in adults, children, and infants*. American Epilepsy Society, Chicago; 2016.

3. Cobb WA, Guiloff RJ, Cast J. Breach rhythm: the EEG related to skull defects. *Electroencephalography and Clinical Neurophysiology*. 1979 Sep 1;**47**(3):251–71.

4. Tsetsou S, Oddo M, Rossetti AO. Clinical outcome after a reactive hypothermic EEG following cardiac arrest. *Neurocritical Care*. 2013 Dec 1;**19**(3):283–6.

5. Thompson SA, Hantus S. Highly epileptiform bursts are associated with seizure recurrence. *Journal of Clinical Neurophysiology*. 2016 Feb 1;**33**(1):66–71.

https://doi.org/10.1017/9781108918923.012 Published online by Cambridge University Press

6. Hofmeijer J, Tjepkema-Cloostermans MC, van Putten MJ. Burst-suppression with identical bursts: a distinct EEG pattern with poor outcome in postanoxic coma. *Clinical Neurophysiology*. 2014 May 1;**125**(5):947–54.

7. Hofmeijer J, van Putten MJ. EEG in postanoxic coma: prognostic and diagnostic value. *Clinical Neurophysiology*. 2016 Apr 1;**127**(4):2047–55.

8. Azabou E, Navarro V, Kubis N, et al. Value and mechanisms of EEG reactivity in the prognosis of patients with impaired consciousness: a systematic review. *Critical Care*. 2018 Dec 1;**22**(1):184.

9. Clemens B, Ménes A. Sleep spindle asymmetry in epileptic patients. *Clinical Neurophysiology*. 2000 Dec 1;**111**(12):2155–9.

10. Huotari AM, Koskinen M, Suominen K, et al. Evoked EEG patterns during burst suppression with propofol. *British Journal of Anaesthesia*. 2004 Jan 1;**92**(1):18–24.

https://doi.org/10.1017/9781108918923.012 Published online by Cambridge University Press